

Lecture-10 [27 August 2026]

Effron-Stein Inequality, McDiarmid's Inequality

Given random variables $X_1, \dots, X_n$ and $g: \mathbb{R}^n \rightarrow \mathbb{R}$, let

$$Z := g(X_1, \dots, X_n).$$

For any $i \in \{1, \dots, n\}$, let

$$Z_i := E[Z | X_1, \dots, X_i], \quad \Delta_i := Z_i - Z_{i-1}.$$

By convention, $Z_0 = E[Z]$.

In the previous lecture, we saw that

$$\text{Var}(Z) = \sum_{i=1}^n \text{Var}(\Delta_i) = \sum_{i=1}^n E[\Delta_i^2].$$

By law of total expectation,
 $E[\Delta_i] = 0$

Theorem (Effron-Stein Inequality)

If $X_1, \dots, X_n$ are mutually independent and $Z = g(X_1, \dots, X_n)$, then

$$\text{Var}(Z) \leq \sum_{i=1}^n E[(Z - E^{(i)}[Z])^2],$$

where

$$E^{(i)}[Z] := E[Z | \underbrace{X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n}_{i^{\text{th}} \text{ RV } X_i \text{ not involved}}]$$

i^{th} RV X_i not involved

Proof of the Effron-Stein inequality

We have $\Delta_i^2 = (Z_i - Z_{i-1})^2 = (E[Z | X_1, \dots, X_i] - E[Z | X_1, \dots, X_{i-1}])^2$

We now claim that

$$E[Z | X_1, \dots, X_{i-1}] = E[E^{(i)}[Z] | X_1, \dots, X_{i-1}]$$

Indeed, for all $x_1, \dots, x_i \in \mathbb{R}$,

$$\mathbb{E}[Y|X=x] = \phi(x)$$

$$\mathbb{E}[Y|X] = \phi(X)$$

$$\mathbb{E}^{(i)}[Z] = \mathbb{E}[g(X_1, \dots, X_n) | X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n]$$

For any $x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n \in \mathbb{R}$,

$$\mathbb{E}[g(X_1, \dots, X_n) | X_1 = x_1, \dots, X_{i-1} = x_{i-1}, X_{i+1} = x_{i+1}, \dots, X_n = x_n]$$

$$= \int g(x_1, \dots, x_{i-1}, x, x_{i+1}, \dots, x_n) f_{X_i | X_1 = x_1, \dots, X_{i-1} = x_{i-1}, X_{i+1} = x_{i+1}, \dots, X_n = x_n}(x) dx$$

$$\begin{aligned}
& E \left[E^{(i)}[Z] \mid X_1 = x_1, \dots, X_i = x_i \right] \\
&= E \left[E[Z \mid X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n] \mid X_1 = x_1, \dots, X_i = x_i \right] \\
&= E \left[E[Z \mid X_1 = x_1, \dots, X_{i-1} = x_{i-1}, X_{i+1}, \dots, X_n] \mid X_1 = x_1, \dots, X_i = x_i \right] \\
&= \int E[Z \mid X_1 = x_1, \dots, X_{i-1} = x_{i-1}, X_{i+1} = x_{i+1}, \dots, X_n = x_n] \\
&\quad \int_{x_{i+1}, \dots, x_n} \left[X_1 = x_1, \dots, X_i = x_i \right] (x_{i+1}, \dots, x_n) dx_{i+1} \dots dx_n \\
&= \int E[Z \mid X_1 = x_1, \dots, X_{i-1} = x_{i-1}, X_{i+1} = x_{i+1}, \dots, X_n = x_n] \int_{x_{i+1}, \dots, x_n} (x_{i+1}, \dots, x_n) dx_{i+1} \dots dx_n \\
&\quad \rightarrow X_{i+1}, \dots, X_n \perp\!\!\!\perp (X_1, \dots, X_i) \\
&= E[Z \mid X_1 = x_1, \dots, X_{i-1} = x_{i-1}].
\end{aligned}$$

Thus, we have

$$\begin{aligned}
\text{Var}(Z) &= \sum_{i=1}^n E[\Delta_i^2] & \Delta_i &= Z_i - Z_{i-1} \\
& & &= Z_i - E^{(i)}[Z] \\
&= \sum_{i=1}^n E \left[\left(E[Z - E^{(i)}[Z] \mid X_1, \dots, X_i] \right)^2 \right] \\
&\stackrel{\text{Conditional Jensen's } f(x)=x^2}{\leq} \sum_{i=1}^n E \left[E[(Z - E^{(i)}[Z])^2 \mid X_1, \dots, X_i] \right] \\
&= \sum_{i=1}^n E[(Z - E^{(i)}[Z])^2].
\end{aligned}$$

Bounded Differences

Definition (Bounded differences property)

A function $g: \mathbb{R}^n \rightarrow \mathbb{R}$ is said to have the bounded differences property if there exist constants $c_1, \dots, c_n \in \mathbb{R}$ such that

$$|g(x_1, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_n) - g(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n)| \leq c_i$$

for all i , for all $x_1, \dots, x_n, y \in \mathbb{R}$.

Example: Let $g: [-1, 1]^n \rightarrow \mathbb{R}$ be defined as

$$g(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n x_i, \quad x_1, \dots, x_n \in [-1, 1].$$

Then,

$$\begin{aligned} & \left| g(x_1, \dots, x_i, x_{i+1}, \dots, x_n) - g(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n) \right| \\ &= \left| \frac{x_i - y}{n} \right| \leq \frac{2}{n}. \end{aligned}$$

Lemma: Let $X_1, \dots, X_n$ be independent and let $g: \mathbb{R}^n \rightarrow \mathbb{R}$ be given.

Let $Z = g(X_1, \dots, X_n)$ and for each $i \in \{1, \dots, n\}$, let

$$E^{(i)}[Z] = E[Z | X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n].$$

Then, for all $i \in \{1, \dots, n\}$,

$$\begin{aligned} & E[(Z - E^{(i)}[Z])^2] = \\ & \frac{1}{2} E \left[\left(\overbrace{g(X_1, \dots, X_{i-1}, \underbrace{X_i}_{Z}, X_{i+1}, \dots, X_n)}^{Z} - \overbrace{g(X_1, \dots, X_{i-1}, \underbrace{Y}_{W_i}, X_{i+1}, \dots, X_n)}^{W_i} \right)^2 \right], \end{aligned}$$

where Y is an IID copy of X_i .

Proof of Lemma

We have

$$\begin{aligned} & E[(Z - W_i)^2 | X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n] \\ &= E[Z^2 | X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n] + E[W_i^2 | X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n] \\ &\quad - 2 E[Z W_i | X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n] \\ &= 2 E[Z^2 | X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n] - 2 \left(E[Z | X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n] \right)^2 \end{aligned}$$

$$= 2 E[Z^2 | X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n] - 2(E^{(i)}[Z])^2$$

$$= 2 E\left[Z^2 - (E^{(i)}[Z])^2 \mid X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n\right]$$

Show this ↪ $= 2 E\left[\left(Z - E^{(i)}[Z]\right)^2 \mid X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n\right].$

Taking expectations on both sides, we get the desired result

As a consequence of the above lemma, if $Z = g(X_1, \dots, X_n)$ and $X_1, \dots, X_n$ are independent, then

$$\text{Var}(Z) \leq \frac{1}{2} \sum_{i=1}^n E\left[\left(g(X_1, \dots, X_i, X_{i+1}, \dots, X_n) - g(X_1, \dots, X_{i-1}, Y, X_{i+1}, \dots, X_n)\right)^2\right]$$

where $X_i \stackrel{\text{iid}}{\sim} Y$.

Furthermore, if $g: \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies BDP with constants $c_1, \dots, c_n$, then

$$\text{Var}(Z) \leq \frac{1}{2} \sum_{i=1}^n c_i^2.$$

Then, for such a function g , we have by Chebyshev's inequality that

$$\mathbb{P}\left(\left|g(X_1, \dots, X_n) - E[g(X_1, \dots, X_n)]\right| \geq \varepsilon\right) \leq \frac{\sum_{i=1}^n c_i^2}{2\varepsilon^2} \quad \forall \varepsilon > 0.$$

Theorem (McDiarmid's inequality)

If $g: \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies the bounded differences property (BDP) with constants $c_1, \dots, c_n \in \mathbb{R}$, and $X_1, \dots, X_n$ are independent, then

$$\mathbb{P}\left(\left|g(X_1, \dots, X_n) - E[g(X_1, \dots, X_n)]\right| \geq \varepsilon\right) \leq 2 \exp\left(-\frac{2\varepsilon^2}{\sum_{i=1}^n c_i^2}\right) \quad \forall \varepsilon > 0.$$

Proof of McDiarmid's inequality

From the previous lemma, we have

$$\text{Var}(Z) \leq \frac{1}{2} \sum_{i=1}^n c_i^2.$$

Recall that

$$\begin{aligned} Z - E[Z] &= \sum_{i=1}^n Z_i - Z_{i-1} & Z_i &= E[g(x_1, \dots, x_n) | x_1, \dots, x_i] \\ &= \sum_{i=1}^n \Delta_i. \end{aligned}$$

We now note that for any $x_1, \dots, x_{i-1} \in \mathbb{R}$,

$$\Delta_i | \{X_1 = x_1, \dots, X_{i-1} = x_{i-1}\} =$$

$$E[g(x_1, \dots, x_n) | x_1, \dots, x_i] - E[g(x_1, \dots, x_n) | x_1, \dots, x_{i-1}] | \{X_1 = x_1, \dots, X_{i-1} = x_{i-1}\}$$

$$= E[g(x_1, \dots, x_{i-1}, x_i, \dots, x_n) | x_1 = x_1, \dots, x_{i-1} = x_{i-1}, x_i] - Z_{i-1}$$

$\hookrightarrow$ function of $x_1, \dots, x_{i-1}$

$$= \int g(x_1, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_n) \underbrace{f_{x_{i+1}, \dots, x_n | x_1 = x_1, \dots, x_{i-1} = x_{i-1}, x_i}(x_{i+1}, \dots, x_n) dx_{i+1} \dots dx_n}_{\substack{f_{x_{i+1}, \dots, x_n}(x_{i+1}, \dots, x_n) \\ \text{because } (X_{i+1}, \dots, X_n) \perp (X_1, \dots, X_i)}} - Z_{i-1}.$$

$$= \int g(x_1, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_n) f_{x_{i+1}, \dots, x_n}(x_{i+1}, \dots, x_n) dx_{i+1} \dots dx_n - Z_{i-1}$$

Clearly,

$$\Delta_i | \{X_1 = x_1, \dots, X_{i-1} = x_{i-1}\} \geq a_i - Z_{i-1}, \quad \Delta_i | \{X_1 = x_1, \dots, X_{i-1} = x_{i-1}\} \leq b_i - Z_{i-1},$$

where

$$b_i = \sup_z \int g(x_1, \dots, x_{i-1}, z, x_{i+1}, \dots, x_n) f_{x_{i+1}, \dots, x_n}(x_{i+1}, \dots, x_n) dx_{i+1} \dots dx_n,$$

$$a_i = \inf_y \int g(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n) f_{x_{i+1}, \dots, x_n}(x_{i+1}, \dots, x_n) dx_{i+1} \dots dx_n.$$

Observe that a_i and b_i are functions of $x_1, \dots, x_{i-1}$. We now claim that

$$b_i - a_i \leq c_i.$$

Indeed, by the bounded differences property of g , keeping $x_1, \dots, x_{i-1}$ fixed,

$$g(x_1, \dots, x_{i-1}, z, x_{i+1}, \dots, x_n) \leq g(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n) + C_i$$

for all $y, z, x_{i+1}, \dots, x_n \in \mathbb{R}$. Therefore, it follows that

$$\begin{aligned} & \int g(x_1, \dots, x_{i-1}, z, x_{i+1}, \dots, x_n) f_{x_{i+1}, \dots, x_n}(x_{i+1}, \dots, x_n) dx_{i+1} \dots dx_n \\ & \leq \int g(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n) f_{x_{i+1}, \dots, x_n}(x_{i+1}, \dots, x_n) dx_{i+1} \dots dx_n + C_i \end{aligned}$$

for all $y, z \in \mathbb{R}$. Taking sup over z on both sides (keeping y fixed), we get

$$b_i \leq \int g(x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n) f_{x_{i+1}, \dots, x_n}(x_{i+1}, \dots, x_n) dx_{i+1} \dots dx_n + C_i$$

for all $y \in \mathbb{R}$. Taking inf over y on both sides, we get

$$b_i \leq a_i + C_i$$

Thus,

$$\Delta_i \mid x_1, \dots, x_{i-1} \in [a_i - z_{i-1}, b_i - z_{i-1}], \quad b_i - a_i \leq C_i$$

From exercise 1(c) of practice problems-03, we have

$$\Delta_i \mid x_1, \dots, x_{i-1} \text{ is } \frac{b_i - a_i}{2} \text{ - sub-Gaussian}$$

$$\Rightarrow \mathbb{E}[e^{t\Delta_i} \mid x_1, \dots, x_{i-1}] \leq \mathbb{E}\left[\exp\left(t^2 \frac{(b_i - a_i)^2}{8}\right) \mid x_1, \dots, x_{i-1}\right] \leq \exp\left(\frac{t^2 c_i^2}{8}\right)$$

Applying Chernoff's technique and noting that

$$\mathbb{E}[e^{t(Z - \mathbb{E}[Z])}] = \mathbb{E}\left[e^{t \sum_{i=1}^n \Delta_i}\right]$$

$$= \mathbb{E}\left[\mathbb{E}\left[e^{t \sum_{i=1}^n \Delta_i} \mid x_1, \dots, x_{n-1}\right]\right]$$

$$= \mathbb{E}\left[e^{t \sum_{i=1}^{n-1} \Delta_i} \cdot \mathbb{E}[e^{t\Delta_n} \mid x_1, \dots, x_n]\right]$$

$$\leq \mathbb{E}\left[e^{t \sum_{i=1}^{n-1} \Delta_i} \cdot e^{t^2 c_n^2 / 8}\right] \dots \leq e^{t^2 \sum_{i=1}^n c_i^2 / 8},$$

we get the desired result.

Application: Largest Eigenvalue of a Symmetric, Random Matrix

Let $\underline{X}$ be an $n \times n$ symmetric matrix with IID entries X_{ij} ,
 $1 \leq i \leq n$, $1 \leq j \leq n$, such that

$$X_{ij} \in [-1, 1] \quad \text{for all } i, j,$$

$$X_{ij} = X_{ji} \quad \text{for all } i, j.$$

If $\underline{X}$ were an $n \times n$ deterministic matrix with all entries equal to 1,
then $\lambda_{\max}(\underline{X}) = 1$.

Two facts: If A is symmetric, then

① All eigenvalues of A are real.

② There exists an orthonormal basis of eigenvectors.

$$\textcircled{3} \quad \lambda_{\max}(A) = \sup_{\underline{v}: \|\underline{v}\|_2=1} \underline{v}^T A \underline{v}$$

Proof of ③: Because A is symmetric, we may express it as

$$A = \sum_{i=1}^n \lambda_i \underline{v}_i \underline{v}_i^T, \quad \{\underline{v}_i\} \rightarrow \text{ONB for } \mathbb{R}^n.$$

$$A \underline{v}_i = \lambda_i \underline{v}_i \quad \forall i.$$

Then, for any $\underline{v} = \sum_{i=1}^n \alpha_i \underline{v}_i$,

$$\underline{v}^T A \underline{v} = \sum_{i=1}^n \lambda_i \alpha_i^2, \quad \|\underline{v}\|_2^2 = \sum_{i=1}^n \alpha_i^2.$$

Thus,

$$\begin{aligned} \sup_{\underline{v}: \|\underline{v}\|_2=1} \underline{v}^T A \underline{v} &= \sup_{\alpha: \sum_{i=1}^n \alpha_i^2=1} \sum_{i=1}^n \lambda_i \alpha_i^2 \\ &= \sup_{\substack{\beta: \beta_i > 0, \\ \sum_{i=1}^n \beta_i = 1}} \sum_{i=1}^n \lambda_i \beta_i = \max_i \lambda_i \\ &= \lambda_{\max}(A). \end{aligned}$$

For a fixed i, j such that $i \leq j$, let $A^{(ij)}$ denote the random matrix

$$A_{k,l}^{(ij)} = X_{k,l} \quad \forall (k,l) \notin \{(i,j), (j,i)\},$$

$$A_{i,j}^{(ij)} = A_{j,i}^{(ij)} = X_{i,j}^{\prime} \leftarrow \text{iid copy of } X_{i,j}.$$

Then,

$$\lambda_{\max}(\underline{X}) - \lambda_{\max}(A^{(ij)}) = \sup_{\underline{v}: \|\underline{v}\|_2=1} \underline{v}^T \underline{X} \underline{v} - \sup_{\underline{v}: \|\underline{v}\|_2=1} \underline{v}^T A^{(ij)} \underline{v}$$

$$\text{Let } \underline{v}^* = \arg\max_{\underline{v}: \|\underline{v}\|_2=1} \underline{v}^T \underline{X} \underline{v}; \quad \text{note that } \|\underline{v}^*\|_2 = 1.$$

Then,

$$\lambda_{\max}(\underline{X}) - \lambda_{\max}(A^{(ij)}) = (\underline{v}^*)^T \underline{X} \underline{v}^* - \sup_{\underline{v}: \|\underline{v}\|_2=1} \underline{v}^T A^{(ij)} \underline{v}$$

$$\leq (\underline{v}^*)^T \underline{X} \underline{v} - (\underline{v}^*)^T A^{(ij)} \underline{v}^*$$

$$= (\underline{v}^*)^T (\underline{X} - A^{(ij)}) \underline{v}^*$$

$$= \sum_{k=1}^n \sum_{l=1}^n v_k^* (X_{k,l} - A_{k,l}^{(ij)}) v_l^*$$

$$= 2 \sum_{k=1}^n \sum_{l=k}^n v_k^* (X_{k,l} - A_{k,l}^{(ij)}) v_l^*$$

$$= 2 v_i^* (X_{i,j} - X_{i,j}^{\prime}) v_j^*$$

$$\leq 4 v_i^* v_j^*$$

$$\Rightarrow \left(\lambda_{\max}(\underline{X}) - \lambda_{\max}(A^{(ij)}) \right)_+ \leq 4 (v_i^* v_j^*)_+.$$

$\swarrow$
 $x_+ = \max\{x, 0\}$ is a non-decreasing
 function

$$\underline{\text{Fact}}: E[(Z - E^{(i)}[Z])^2] = 2 E[(Z - E^{(i)}[Z])_+^2] = 2 E[(Z - E^{(i)}[Z])_-^2].$$

We then have

$$\begin{aligned}\text{Var}(\lambda_{\max}(\underline{X})) &\leq \sum_{i=1}^n \sum_{j=i}^n E\left[\left(\lambda_{\max}(\underline{X}) - \lambda_{\max}(A^{(ij)})\right)_+^2\right] \\&\leq \sum_{i=1}^n \sum_{j=i}^n 16 E[(v_i^* v_j^*)^2] \\&\leq 16 \cdot E\left[\sum_{i=1}^n \sum_{j=1}^n (v_i^*)^2 (v_j^*)^2\right] \\&= 16 \cdot \left(E[\|\underline{v}^*\|_2^2]\right)^2 \quad (\|\underline{v}^*\|_2^2 = 1) \\&= 16,\end{aligned}$$

irrespective of n .
